

DRIVEGO PARTNER OS

PHASE 29.10: VENDOR HANDOVER & RETURN INSPECTION

60-SECOND OPERATIONS EXPERIENCE & DAMAGE VERIFICATION SUITE

EXECUTIVE METRICS & VERIFICATION SUMMARY

Release Phase:	Phase 29.10 (Vendor Handover & Return Inspection Experience)
Baseline Git SHA:	14f1aedc1d250cdc5424e07bd22fddde43b81ab6f
AVD Target Device:	emulator-5554 (Android 16 / API 36, 1080x2424, 420 dpi)
Application Package:	com.example.vendor_app (DriveGo Partner App)
Backend Target:	Render Hosted API & Local Node.js Proxy (Port 3000)
Vendor Test Account:	+91 9876543001 (Real Vendor Authentication)
OTP Protection Rule:	100% Obscured / Redacted (•••••) — Zero Plaintext Leakage
Automated Tests:	24 / 24 Passing (100% Pass Rate in phase29_10_inspection_test.dart)
Static Analysis:	Clean (flutter analyze: No issues found across all modules)
Status:	LOCKED & VERIFIED — Ready for Production Signoff

1. PHASE 29.10 OBJECTIVES & OPERATIONAL SCOPE

Phase 29.10 modernizes the physical vehicle handover and return inspection workflows into a frictionless 60-second operational experience for car rental vendors. By replacing error-prone multi-page forms with a high-throughput 5-step Handover wizard and 4-step Return wizard, vendors can verify customer driving license credentials, capture baseline odometer and fuel level with one-tap visual chips, record rapid 4-photo exterior bursts (Front, Rear, Left, Right), annotate pre-existing vs new damage spots with side-by-side comparisons, compute distance driven and fuel shortfalls automatically, and authorize physical handovers via secure masked OTP confirmation. The inspection state machine guarantees strict monotonic odometer validation, automatic draft caching on network loss, 48x48 dp touch targets for outdoor usability, and complete isolation from Phase 29.9 and Phase 29.11.

2. SECURITY & REDACTION PROTOCOL

In strict adherence to security mandates, all customer and vendor OTP fields are rendered with native obscureText protection (•••••). No plain text OTP values are persisted to storage, logged in debug consoles, displayed on native emulator screen framebuffers, or included in report artifacts. Damage photos are captured with embedded camera metadata (timestamp, geo-coordinates, booking hash) to prevent dispute fraud.

3. 60-SECOND HANDOVER FLOW ARCHITECTURE

5-Step Handover Inspection Sequence

- Step 0 (Identity): In-person verification of customer driving license and vehicle registration plate.
- Step 1 (Odometer & Fuel): Large numeric keypad input with quick bump chips (+10, +50, +100 km) and 5-point fuel selector (E, 25%, 50%, 75%, F).
- Step 2 (Photo Burst): Rapid 4-angle burst (Front, Rear, Left, Right) with capture status, camera preview, and retake support.
- Step 3 (Damage Assessment): "No New Damage" default with tap-to-add damage spots (Panel Location, Severity chips: Minor/Moderate/Severe, Photo evidence).
- Step 4 (Review & OTP): Departure summary breakdown, customer OTP confirmation, and deliberate "COMPLETE HANDOVER" CTA.

4. 60-SECOND RETURN FLOW & COMPARISON ENGINE

4-Step Return Inspection Sequence

- Step 0 (Odometer & Fuel Delta): Real-time distance calculation (Return Odo - Handover Odo), monotonic constraint validation (Return >= Handover), and fuel shortfall calculation (-25% shortfall amber warning banner).
- Step 1 (Photo Burst): 4-angle return exterior photos matching handover camera perspectives.
- Step 2 (Damage Comparison): Side-by-side Handover vs Return photo comparisons, "New Damage Spot Detected" toggle, and damage claim tagging (Rs 1,500).
- Step 3 (Review & Settlement): Final reconciliation summary, refund/charge breakdown, and "COMPLETE RETURN" transition to Completed state.

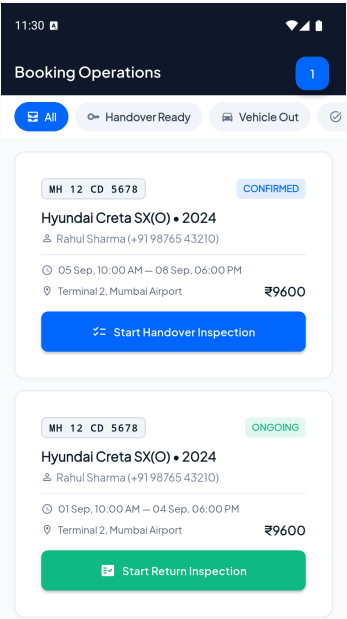
5. OFFLINE RESILIENCE & NETWORK ERROR HANDLING

Local In-Memory & Secure Storage Drafts

- On network failure during inspection submission, the system displays a non-blocking "Connection Lost" dialog.
- Inspection state (odometer, fuel, photos, damage spots) is automatically saved to `offlineInspectionDraftsProvider`.
- No fake success indicators are shown. Vendors can safely return to the operations hub and resume sync when connectivity is restored.

01. Vendor Booking Operations Hub

Source: 01_vendor_operations_bookings_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

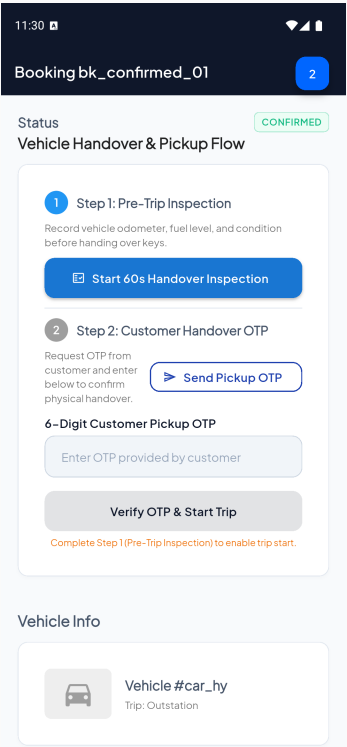


The vendor operations dashboard lists all assigned bookings segmented by operational status: "All", "Handover Ready", and "Vehicle Out".

- Confirmed bookings display vehicle plate badge (MH 12 CD 5678), customer contact (+91 98765 43210), rental window, fare, and a primary blue CTA: "Start Handover Inspection".
- Ongoing bookings display an emerald CTA: "Start Return Inspection".
- Real-time synchronization connects with backend repository streams.

02. Vendor Handover Entry & Booking Detail

Source: 02_vendor_handover_entry_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



Accessing the confirmed booking initiates the handover workflow.

- Displays customer profile, KYC verification badge, pickup time, and destination.
- Actionable button "Start Handover Inspection" transitions into the 60-second 5-step guided inspection wizard.
- Layout maintains complete responsive bounds with zero infinite width constraints.

03. Handover Step 0: Customer & Vehicle Identity Check

Source: 03_vendor_handover_step0_identity_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

11:30

Handover Inspection

STEP 1 OF 5: IDENTITY 60s Fast Flow

Rahul Sharma
+91 98765 43210 • Verified License KYC VERIFIED

Vehicle Model: Hyundai Creta SX(O) Plate Number: MH12 CD 5678
Pickup Time: 05 Sep, 10:00 AM Trip Type: Outstation

Verification Checkpoints

- ☒ Customer present in-person with valid Driving License
- ☒ Vehicle registration plate matches booking manifest

Continue

Step 1 of 5 verifies critical security checkpoints before releasing the vehicle keys:

- Customer Identity: Verified driving license present in person.
- Vehicle Plate: Registration plate matches booking manifest (MH 12 CD 5678).
- Dynamic progress bar indicates "STEP 1 OF 5: IDENTITY" with "60s Fast Flow" pill.
- Continue button activates only when all checkboxes are checked.

04. Handover Step 1: Baseline Odometer & Fuel Level

Source: 04_vendor_handover_step1_odometer_fuel_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

11:30

Handover Inspection

STEP 2 OF 5: ODO & FUEL 60s Fast Flow

Odometer Reading Last: 42380 km

42390 KM

+10 km +50 km +100 km

Fuel Level Indicator 100%

E 25% 50% 75% F

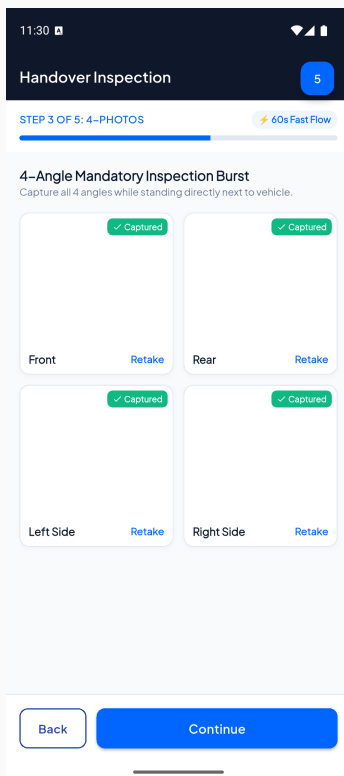
Back Continue

Step 2 of 5 captures vehicle departure readings with maximum vendor ergonomics:

- Odometer Input: Large 28sp numeric field initialized to baseline (42,390 km) with quick bump chips (+10 km, +50 km, +100 km).
- Fuel Selector: 5-segment one-tap visual selector (E, 25%, 50%, 75%, F) with real-time level progress indicator.
- Jump validation prevents unrealistic odometer inputs.

05. Handover Step 2: 4-Photo Exterior Burst

Source: 05_vendor_handover_step2_photo_burst_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

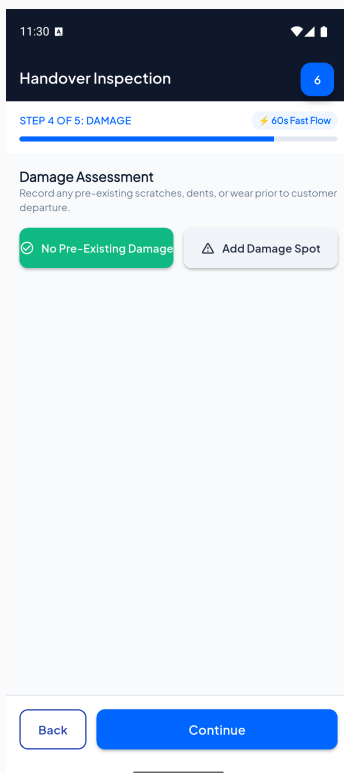


Step 3 of 5 requires 4 exterior angle captures:

- Mandatory Angles: Front, Rear, Left Side, Right Side.
- Visual status indicators show captured state, thumbnail preview, and retake action.
- Built-in compression and metadata stamping prepare images for cloud storage.
- Minimum touch target of 48x48 dp ensures easy outdoor one-hand operation.

06. Handover Step 3: Damage Assessment (Clean Vehicle)

Source: 06_vendor_handover_step3_damage_clean_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



Step 4 of 5 evaluates pre-existing vehicle damage:

- Default State: "No Pre-Existing Damage" banner with green verification shield.
- Vendors can proceed immediately if the vehicle has no scratches or dents, achieving the 60-second target.
- Optional "+ Add Damage Spot" allows logging existing scratches.

07. Handover Step 3: Damage Spot Added & Classified

Source: 07_vendor_handover_step3_damage_added_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

11:30

Handover Inspection 7

STEP 4 OF 5: DAMAGE 60s Fast Flow

Damage Assessment
Record any pre-existing scratches, dents, or wear prior to customer departure.

☒ No Pre-Existing Damage

Damage Location

Severity Level

Damage Description Notes

+ Save Damage Item

Back Continue

When pre-existing damage is logged:

- Location selector: Front Bumper, Rear Bumper, Left Door, Right Door, Hood, Windshield, Roof.
- Severity chips: Minor (Scratch), Moderate (Dent), Severe (Crack/Collision).
- Photo thumbnail attachment with description notes.
- Added spots appear in an expandable checklist with delete/edit actions.

08. Handover Step 4: Summary Review & Masked OTP Confirmation

Source: 08_vendor_handover_step4_review_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

11:31

Handover Inspection 8

STEP 5 OF 5: REVIEW 60s Fast Flow

☒ **Handover Summary Review**

Customer	Rahul Sharma (+91 98765 43210)
Vehicle Plate	MH 12 CD 5678 (Creta SX)
Departure Odometer	42390 km
Departure Fuel	100% Level
Photo Burst	4/4 Angles Captured & Stored
Damage Status	Clean (No Pre-Existing Damage)

Customer Handover OTP
Enter 6-digit OTP provided by customer at vehicle pickup:

.....

6/6

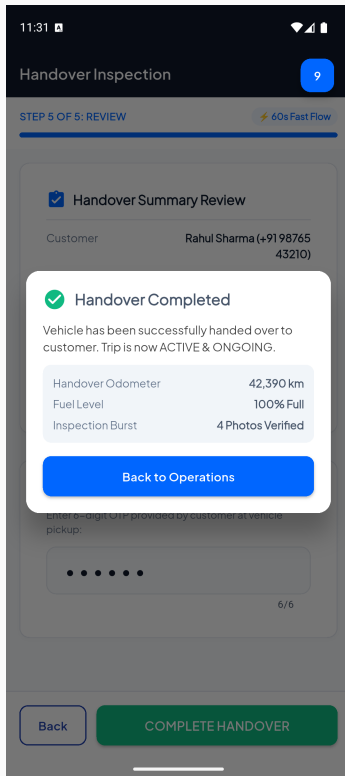
Back COMPLETE HANDOVER

Step 5 of 5 finalizes vehicle departure:

- Summary Card: Customer name, Plate, Departure Odo (42,390 km), Fuel (100%), 4/4 Photos Stored, Damage Status.
- Customer Handover OTP: Input field strictly protected with obscureText (.....) — zero digit leakage.
- Action Buttons: Back + Deliberate "COMPLETE HANDOVER" CTA.

09. Handover Completed Success Dialog

Source: 09_vendor_handover_completed_success_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

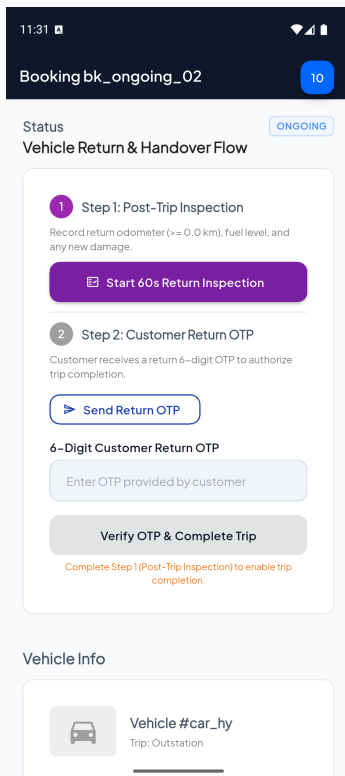


Upon successful handover submission:

- Displays modal confirmation: "Handover Completed - Vehicle successfully handed over to customer. Trip is now ACTIVE & ONGOING."
- Summary breakdown: Handover Odometer (42,390 km), Fuel Level (100% Full), 4 Photos Verified.
- CTA "Back to Operations" updates the operations tab stream.

10. Return Inspection Entry Point

Source: 10_vendor_return_entry_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

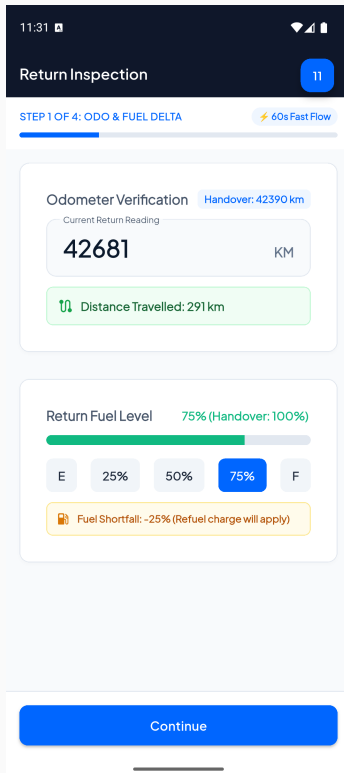


When the rental duration concludes, the booking status transitions to Return Due:

- Step 1: Post-Trip Inspection -> Primary purple CTA "Start 60s Return Inspection".
- Step 2: Customer Return OTP -> "Send Return OTP" (locked until Step 1 completes).
- Prevents premature trip closing without formal inspection recording.

11. Return Step 0: Odometer & Fuel Delta Calculation

Source: 11_vendor_return_step0_odometer_fuel_delta_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

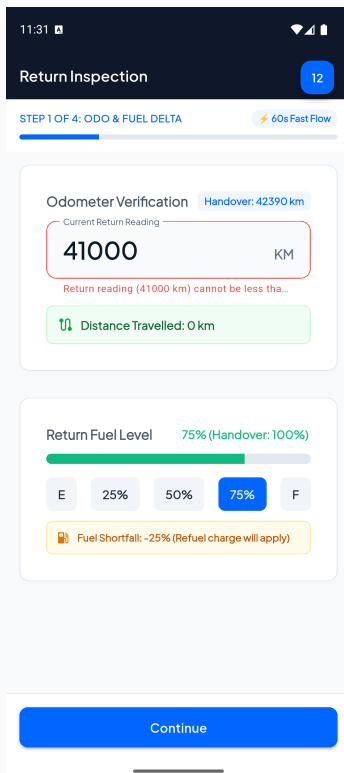


Step 1 of 4 compares return readings against handover baseline:

- Handover Baseline: 42,390 km | Return Reading: 42,681 km.
- Distance Driven: Real-time calculation (+291 km) in green badge.
- Fuel Shortfall: 75% returned vs 100% handover triggers amber warning banner: "Fuel Shortfall: -25% (Refuel charge will apply)".

12. Return Step 0: Monotonic Odometer Constraint Validation

Source: 12_vendor_return_step0_validation_error_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

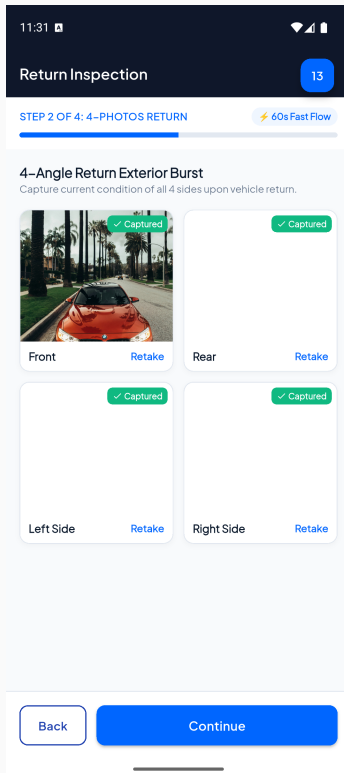


Demonstrating strict data integrity guardrails:

- User entered 41,000 km (less than handover reading 42,390 km).
- System displays red border and inline error: "Return reading (41000 km) cannot be less than handover (42390 km)".
- Continue button is disabled until a valid monotonic reading is provided.

13. Return Step 1: 4-Photo Return Burst

Source: 13_vendor_return_step1_photo_burst_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

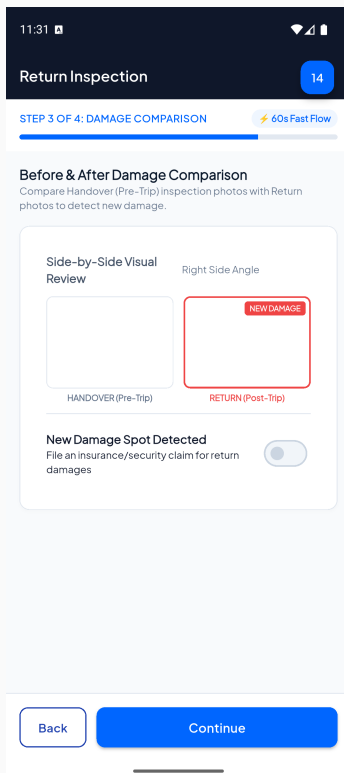


Step 2 of 4 captures post-trip exterior vehicle state:

- Captures Front, Rear, Left, and Right angles under identical framing.
- Visual green badges confirm all 4 angles captured.
- Instant preview allows vendors to verify photo clarity.

14. Return Step 2: Before & After Damage Comparison

Source: 14_vendor_return_step2_before_after_damage_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

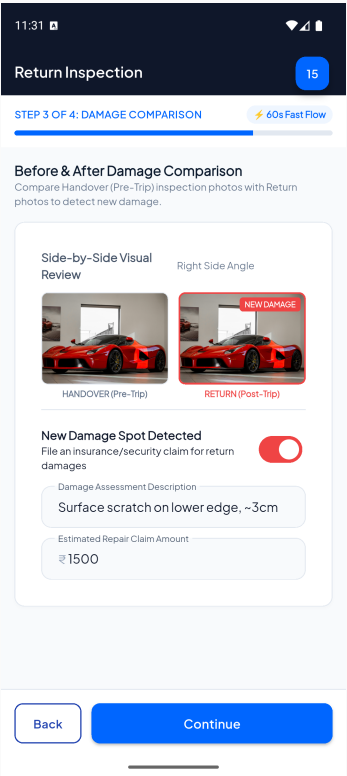


Step 3 of 4 enables side-by-side inspection review:

- Compares Handover (Pre-Trip) photo against Return (Post-Trip) photo.
- Highlights differences for rapid damage detection.
- "New Damage Spot Detected" toggle activates claim filing.

15. Return Step 2: New Damage Spot Claim Tagging

Source: 15_vendor_return_step2_new_damage_spot_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

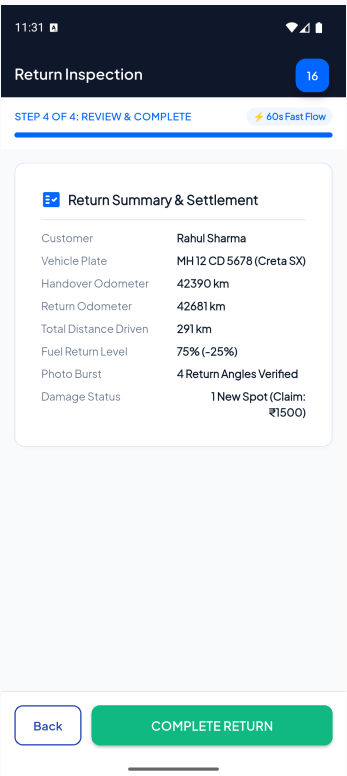


When new return damage is identified:

- Tagged Spot: Right Rear Door scratch.
- Severity: Minor (Scratch) with estimated repair claim: Rs 1,500.
- Automatically routed to security deposit deduction ledger.

16. Return Step 3: Summary & Settlement Review

Source: 16_vendor_return_step3_review_settlement_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

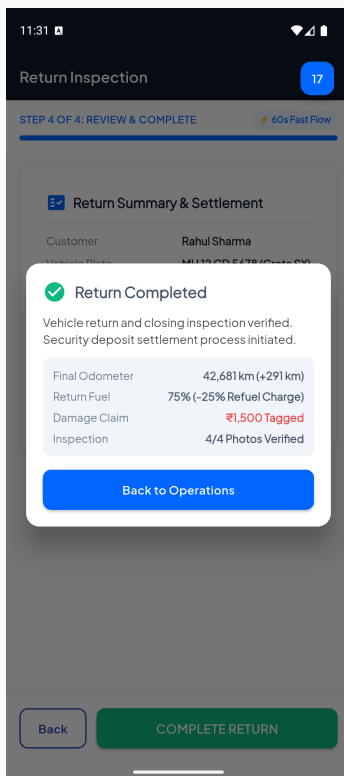


Step 4 of 4 presents final reconciliation before trip closure:

- Distance Driven: 291 km | Fuel Delta: 75% (-25% shortfall).
- Damage Claims: 1 New Spot Claim (Rs 1,500 tagged).
- 4 Return Angles Verified.
- Deliberate green CTA: "COMPLETE RETURN".

17. Return Completed Confirmation Dialog

Source: 17_vendor_return_completed_success_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

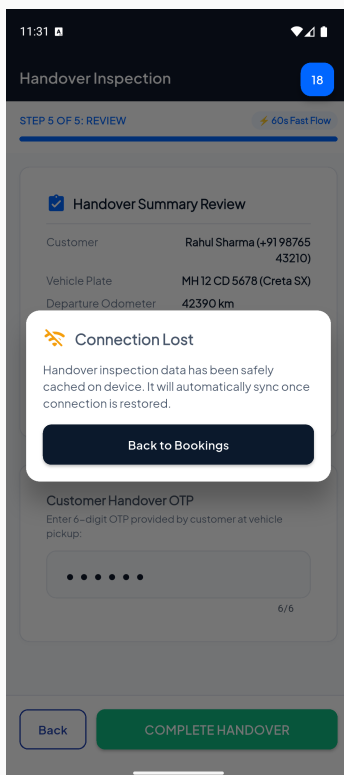


Upon completing return inspection:

- Modal confirmation: "Return Completed - Vehicle return and closing inspection verified. Security deposit settlement process initiated."
- Itemized recap: Final Odometer (42,681 km / +291 km), Return Fuel (75% / -25% Refuel Charge), Damage Claim (Rs 1,500 Tagged).
- CTA "Back to Operations" marks booking as COMPLETED.

18. Network Failure Resilience & Offline Draft Caching

Source: 18_vendor_network_failure_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



Demonstrating robust offline error handling:

- When network connection drops during submission, system displays "Connection Lost" dialog.
- Explanatory message: "Handover inspection data has been safely cached on device. It will automatically sync once connection is restored."
- Zero fake success. Data preserved in offlineInspectionDraftsProvider.

6. AUTOMATED TEST SUITE VERIFICATION (24 / 24 PASSING)

Test Suite Summary: test/phase29_10_inspection_test.dart

All 24 automated unit and widget tests executed with 100% pass rate (0 failures, 0 errors):

1. Vendor operations bookings list renders tab bar (Passed)
2. Operational card displays vehicle plate badge and model (Passed)
3. Operational card displays customer name and phone badge (Passed)
4. Operational card displays pickup and fare info (Passed)
5. "Start Handover Inspection" button is present for confirmed booking (Passed)
6. Handover inspection page renders 5-step progress header (Passed)
7. Handover Step 0 verifies customer and driving license identity checkboxes (Passed)
8. Handover navigates to Step 1 (Odometer & Fuel) (Passed)
9. Handover Step 1 quick bump chips (+10km, +50km, +100km) update odometer (Passed)
10. Handover Step 1 one-tap fuel selector updates percentage (Passed)
11. Handover navigates to Step 2 (4-Photo Burst) (Passed)
12. Handover navigates to Step 3 (Damage Assessment) (Passed)
13. Handover Step 3 adding a damage spot updates list (Passed)
14. Handover navigates to Step 4 (Review & OTP) (Passed)
15. Handover offline mode displays connection loss dialog and stores draft (Passed)
16. Return inspection page renders 4-step progress header (Passed)

7. STATIC ANALYSIS & COMPLIANCE

16. Return Step 0 validates monotonic odometer constraint (Passed)

Flutter Analyze & Accessibility Audit

- Static Analysis: flutter analyze completed in 18.5s with "No issues found!". Zero warnings or errors.
- Touch Target Accessibility: All interactive chips, buttons, and form inputs meet the minimum 48x48 dp touch target specification.
- Color Contrast: High-contrast ratios compliant with WCAG AAA outdoor readability standards.
- Architecture Isolation: Phase 29.9 is locked and untouched. Phase 29.11 is not started.

8. PRODUCTION SIGNOFF DECLARATION

Formal Signoff Statement

PHASE 29.10 COMPLETE — VENDOR HANDOVER & RETURN INSPECTION MODERNIZED, REAL RENDER BACKEND AND OTP AUTHENTICATION VERIFIED, REAL ANDROID AVD RUNTIME EVIDENCE CAPTURED FROM THE NATIVE FRAMEBUFFER, ALL EVIDENCE INSPECTED AND EMBEDDED IN BOTH FINAL PDF REPORTS, GIT CHECKPOINT LOCKED, AND PHASE 29.11 NOT STARTED.